

Protocol Architecture, TCP/IP, and Internet- Based Applications

CHAPTER 2

The Need for a Protocol Architecture

1) The source must **activate** communications path or **inform** the network of destination

2) The source system must **check** the destination is **prepared** to receive data

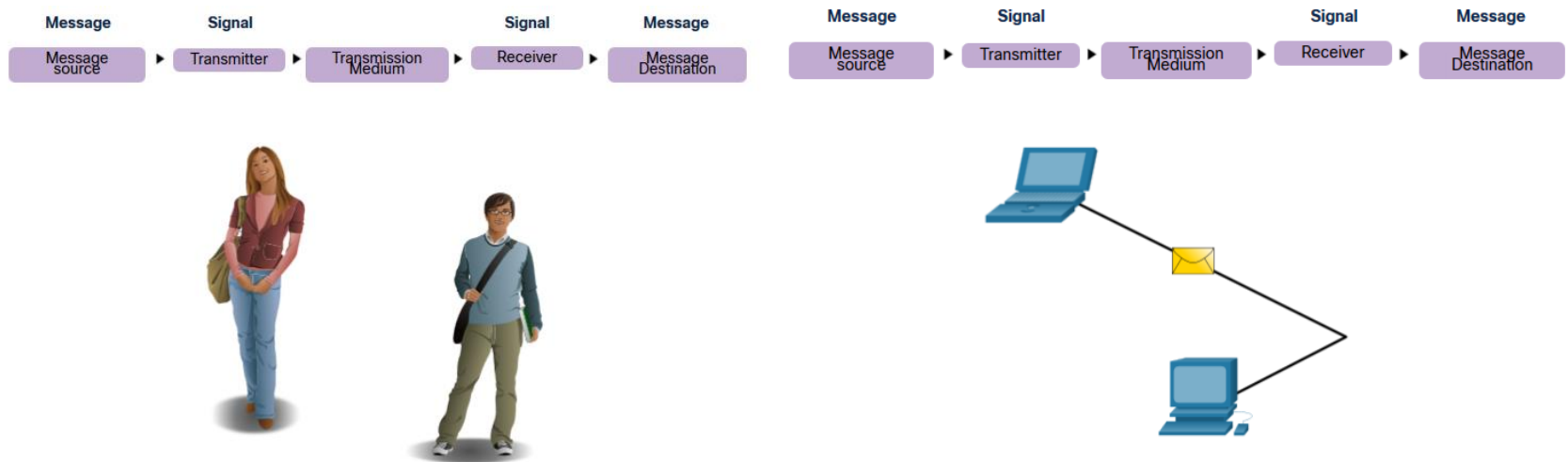
**To transfer data
several tasks must be
performed:**

3) The **file transfer application** on the source must check destination **file management** system is prepared to accept and store the file for this user

4) May need file format translation

Communications Protocols

- All communications are **governed** by protocols.
- Protocols are **the rules that communications will follow**.
- These rules will **vary depending on the protocol**.



Rule Establishment (Cont.)

Protocols must account for the following requirements:

- An identified sender and receiver
- Common language and grammar
- Speed and timing of delivery
- Confirmation or acknowledgment requirements

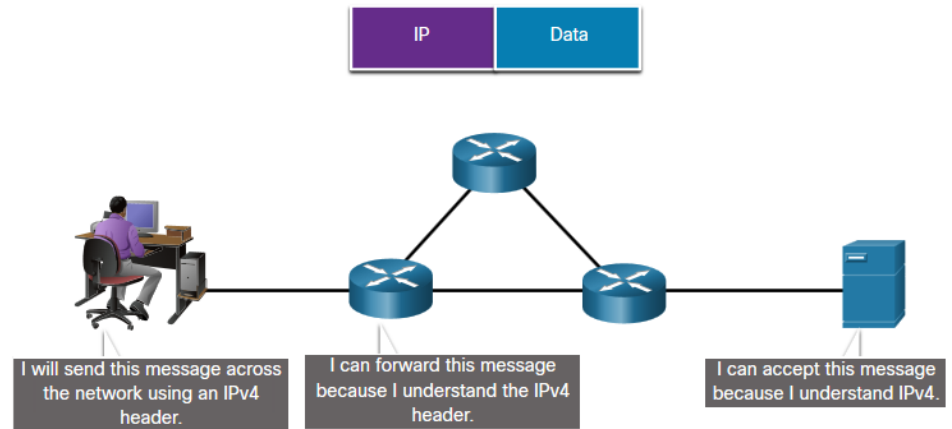
Network Protocol Requirements

Common computer protocols must agree and include the following requirements:

- Message encoding
- Message formatting and encapsulation
- Message size
- Message timing
- Message delivery options

Network Protocol Functions

- Devices use agreed-upon protocols to communicate.
- Protocols **may have one or more functions.**

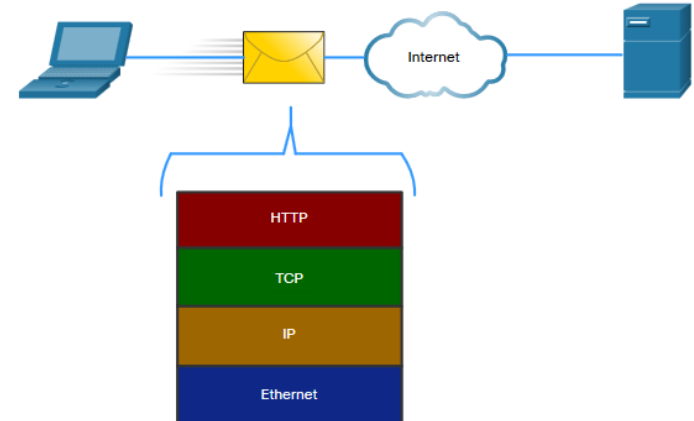


Function	Description
Addressing	Identifies sender and receiver
Reliability	Provides guaranteed delivery
Flow Control	Ensures data flows at an efficient rate
Sequencing	Uniquely labels each transmitted segment of data
Error Detection	Determines if data became corrupted during transmission
Application Interface	Process-to-process communications between network applications

Protocol Interaction

- Networks require the use of several protocols.
- Each protocol has its own function and format.

Protocol
Hypertext Transfer Protocol (HTTP)
Transmission Control Protocol (TCP)
Internet Protocol (IP)
Ethernet



Elements of a Protocol

A protocol is a set of rules that allow peer layers to communicate
The key features of a protocol are:

Syntax

Format of data
blocks

Semantics

Control
information
and error
handling

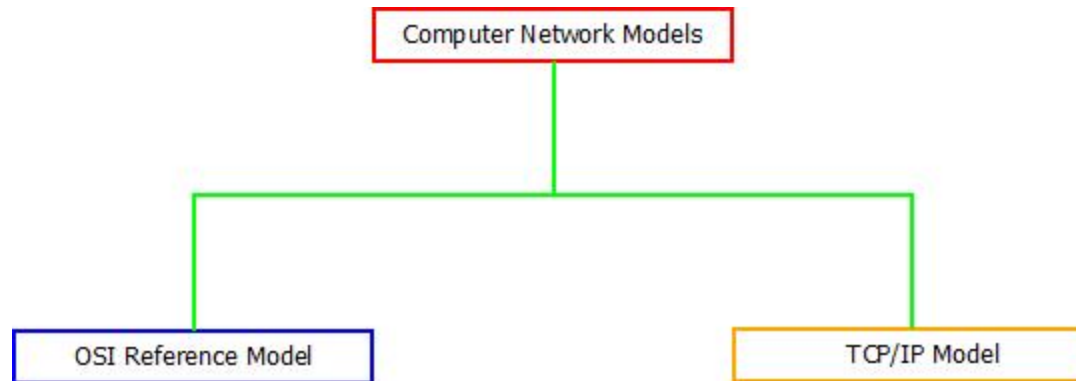
Timing

Speed
matching and
sequencing

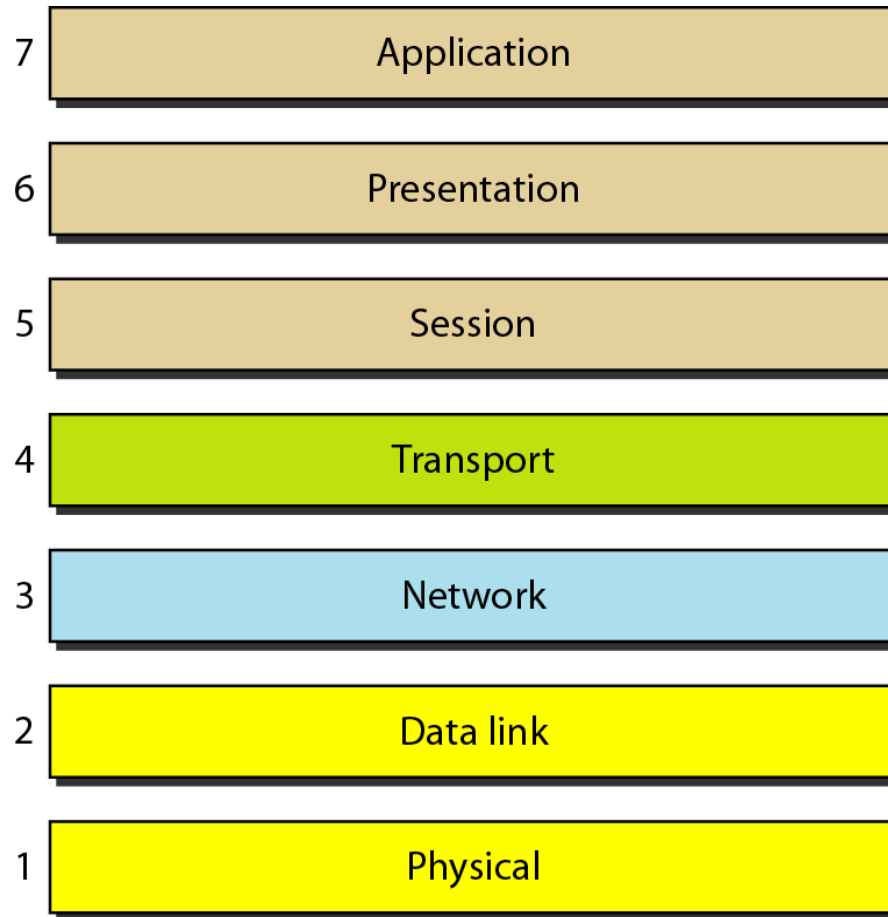
Elements of a Protocol

- **Syntax:** structure or format of data
 - Example: 8-bits address of sender, 8-bits address of receiver
- **Semantics:** meaning of each section of bits
 - Example: Does the address is a route to be taken or the final destination of the message
- **Timing:** when data should be sent and how fast they can be sent
 - Example: sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps ⇒ overload and data loose

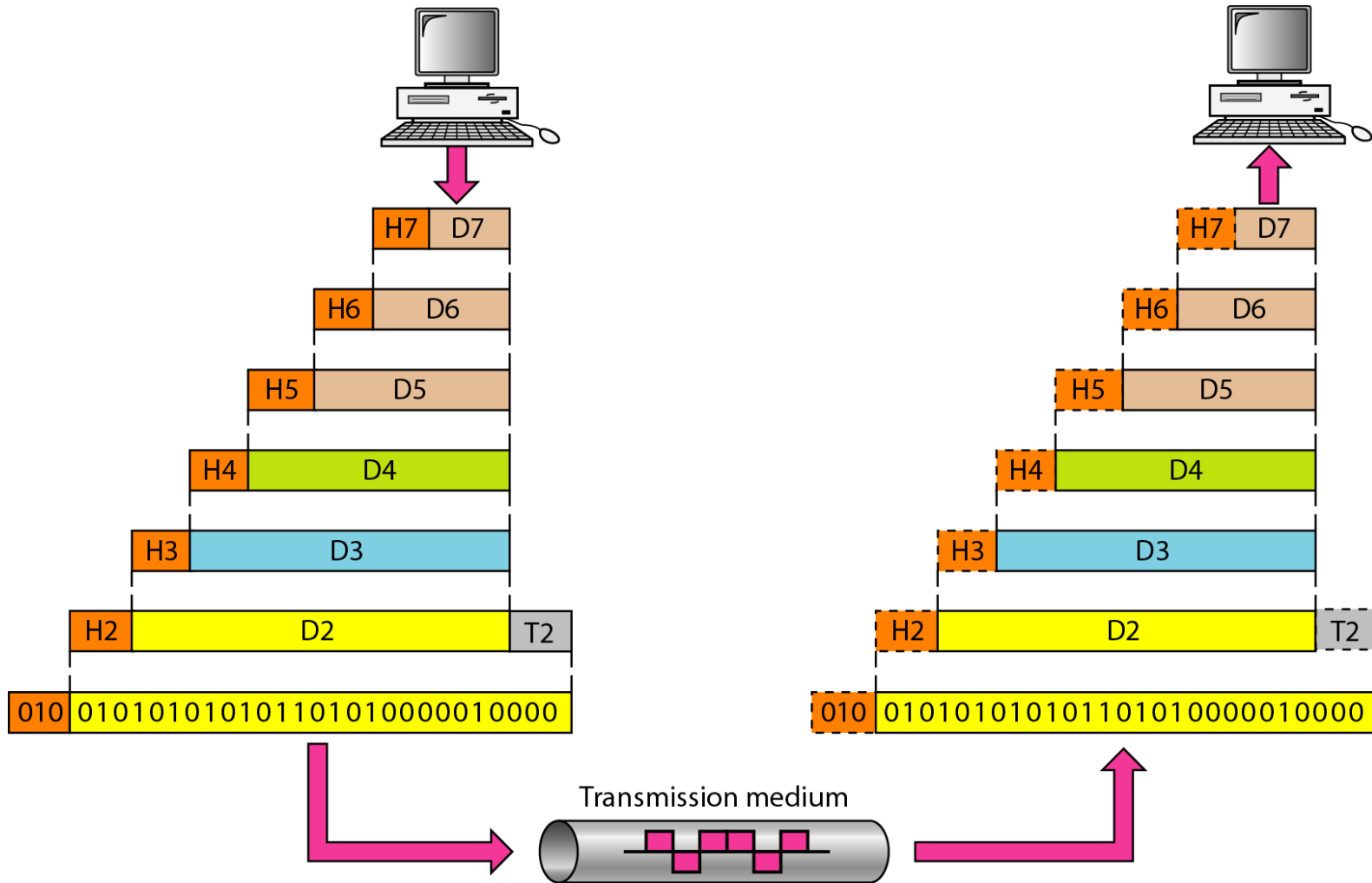
Network Models



Seven layers of the OSI model

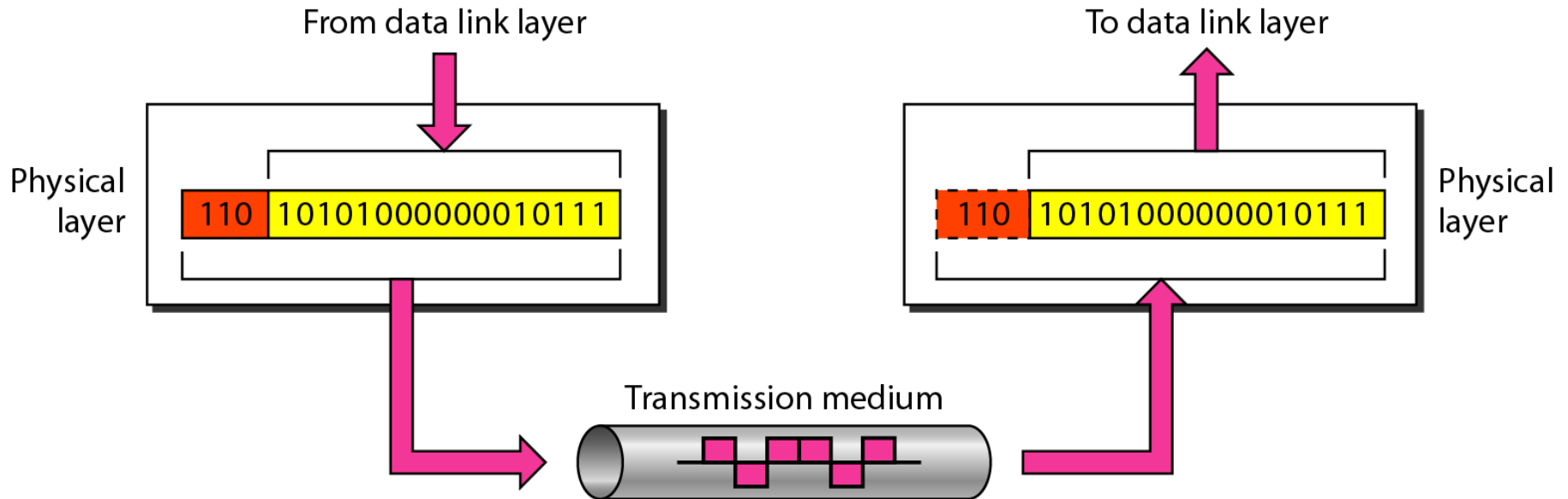


An exchange using the OSI model

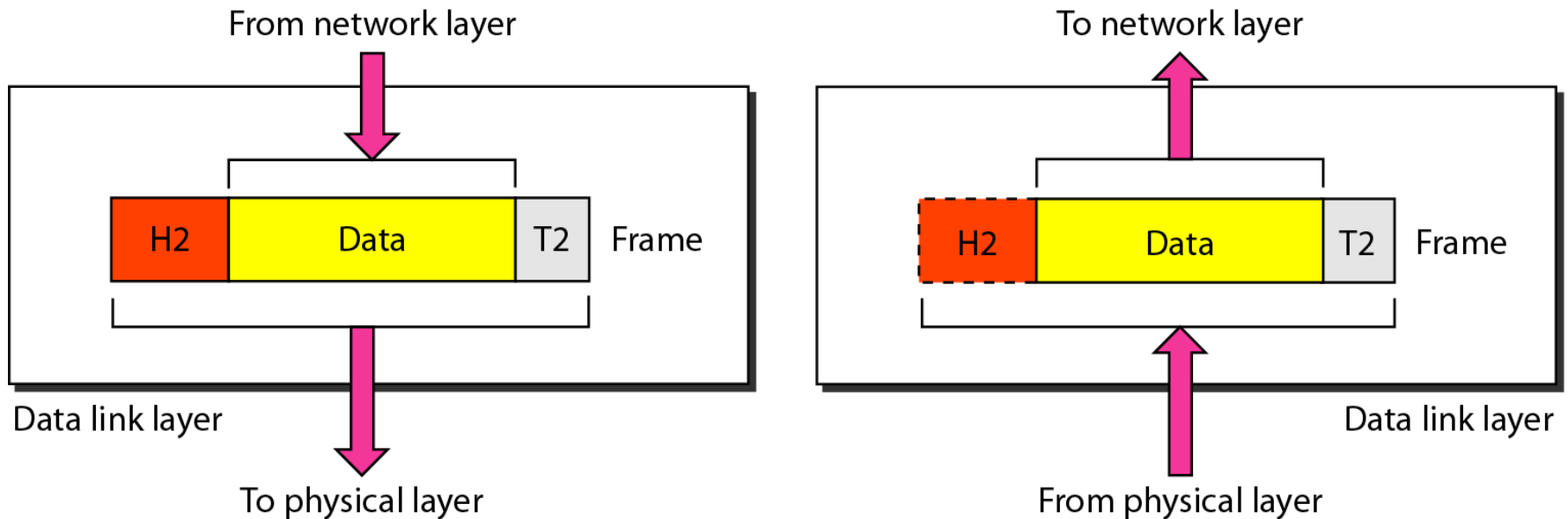


Physical layer

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

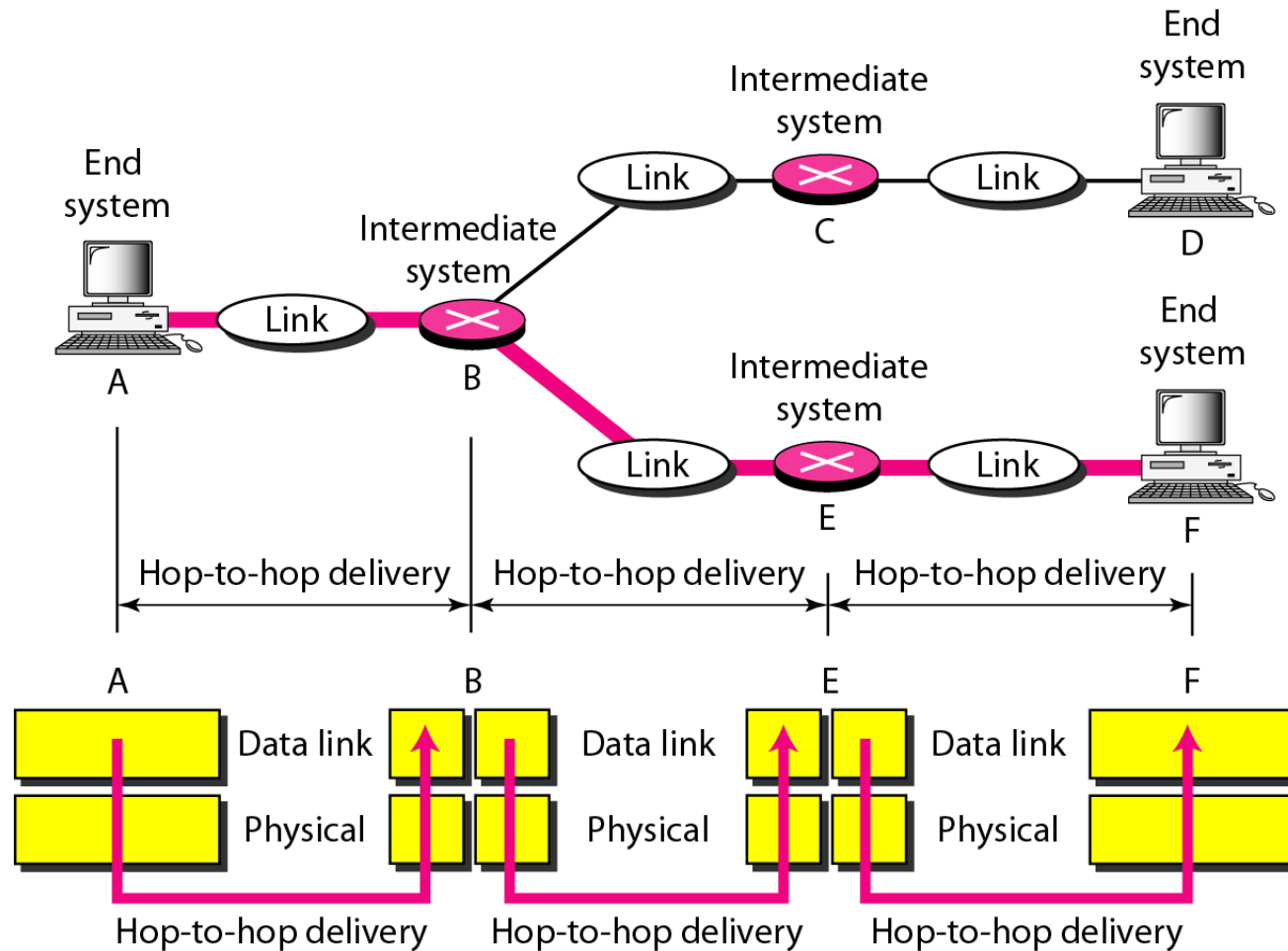


Data link layer

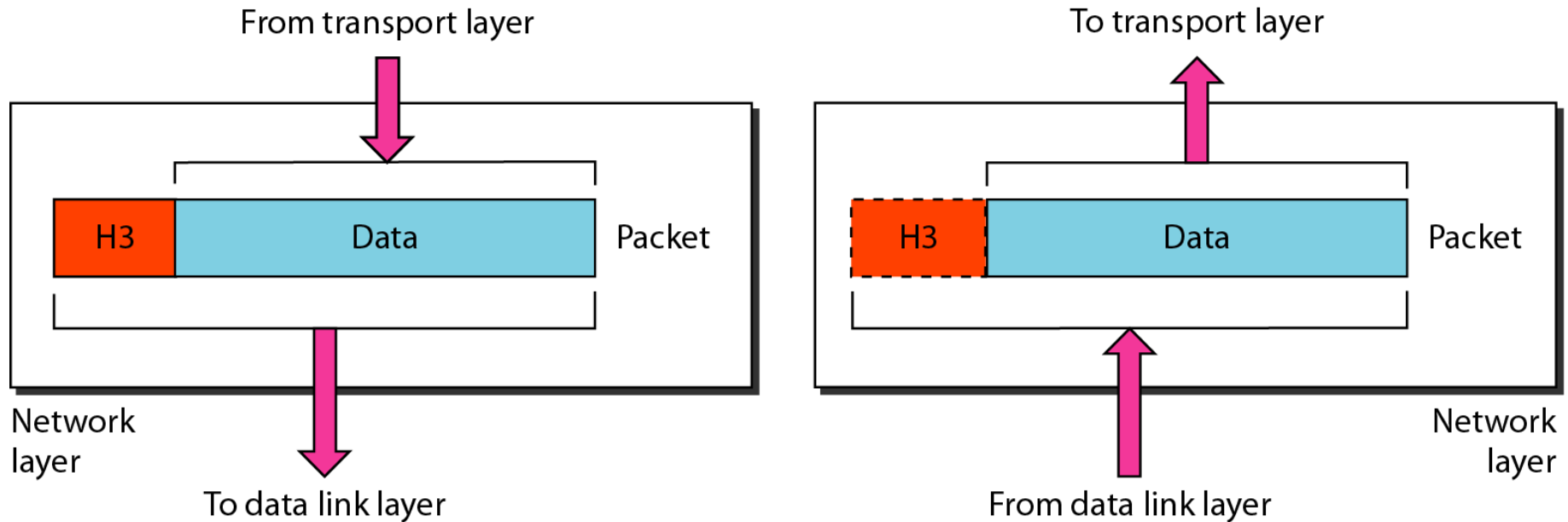


The data link layer is responsible for moving frames from one hop (node) to the next.

Hop-to-hop delivery

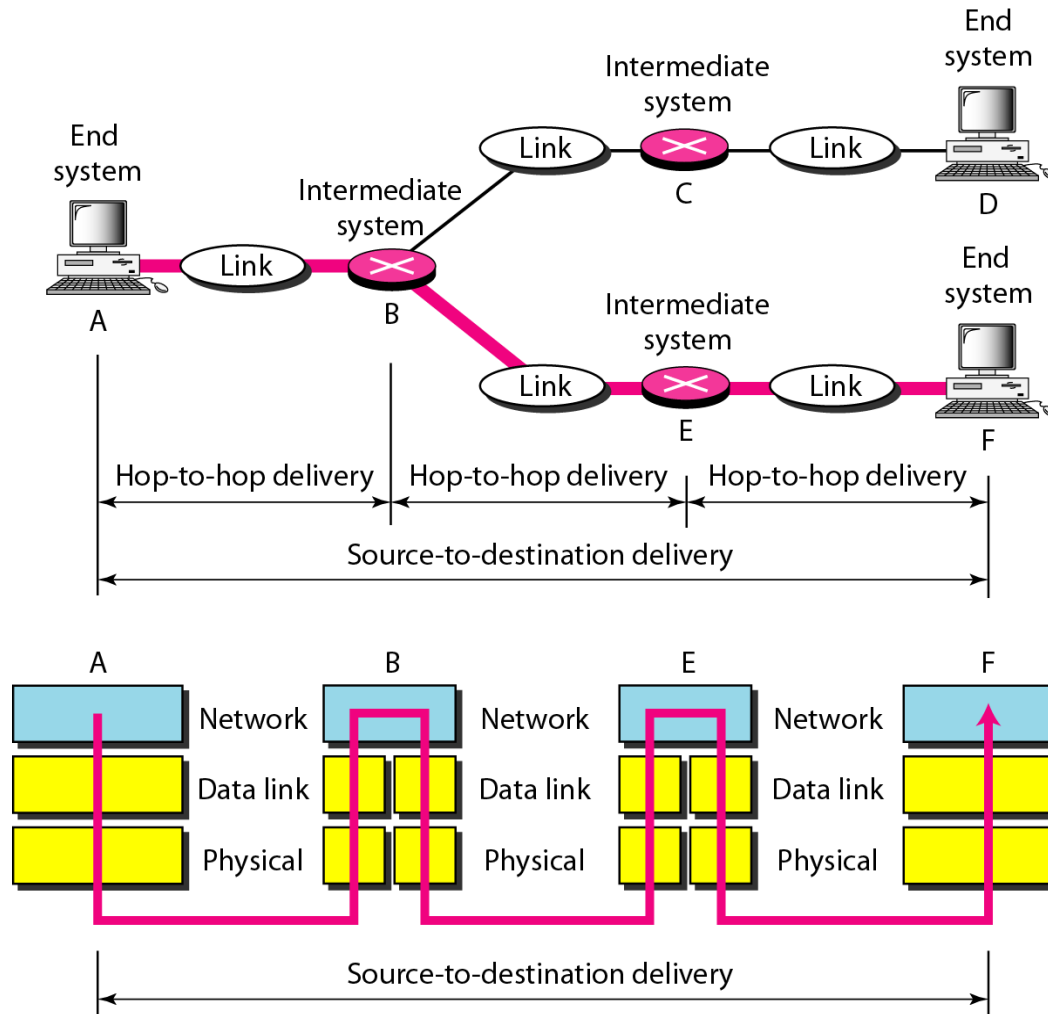


Network layer



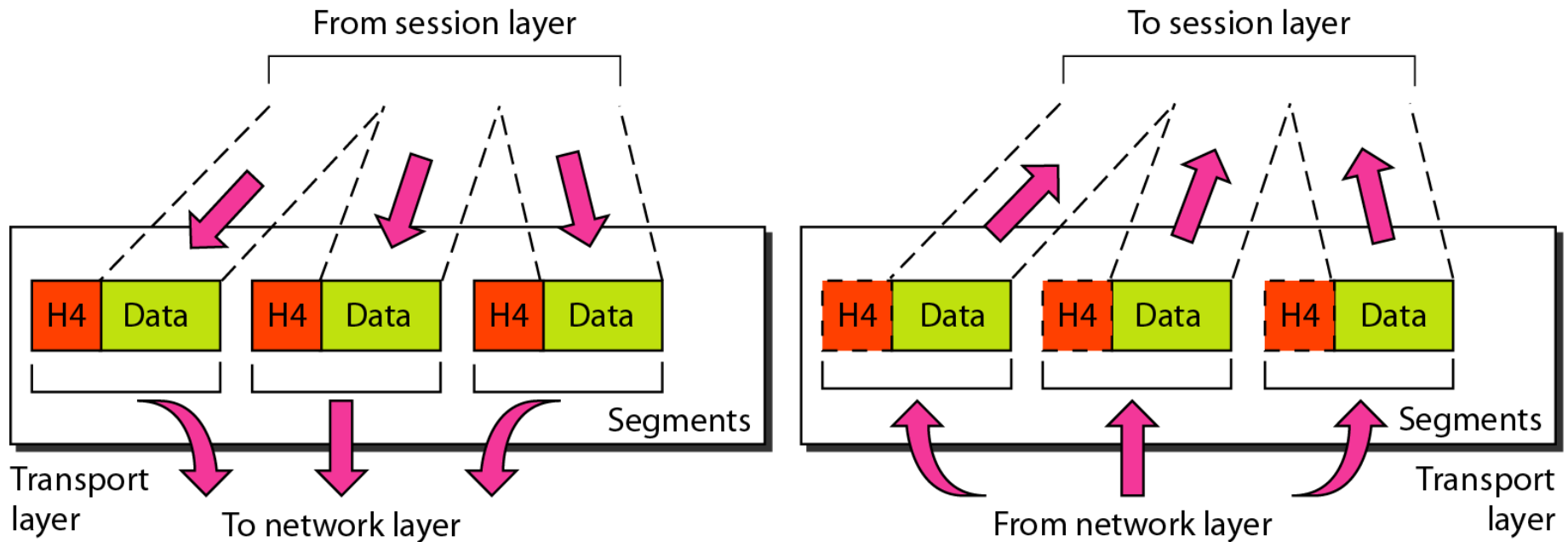
The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Source-to-destination delivery



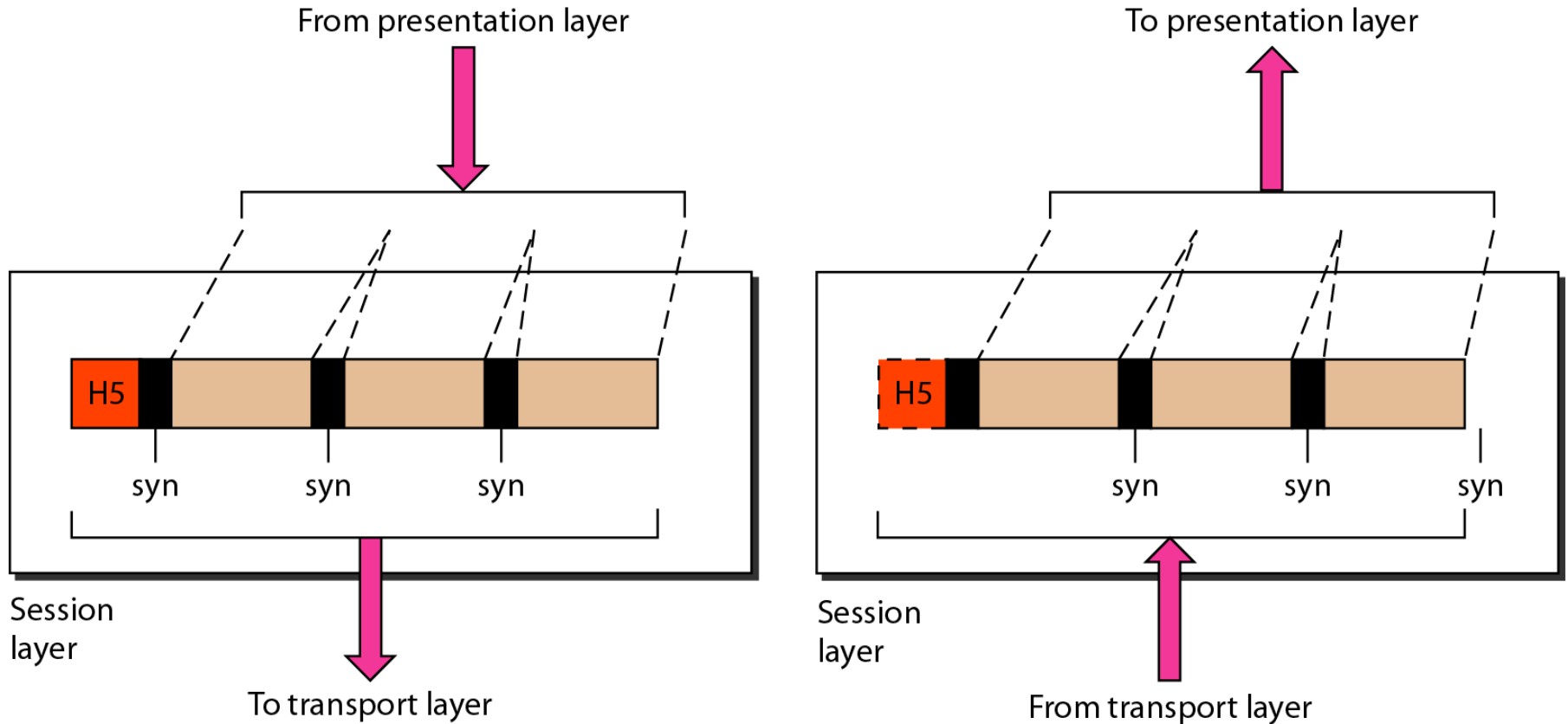
Transport layer

The transport layer is responsible for the delivery of a **message** from one process to another.

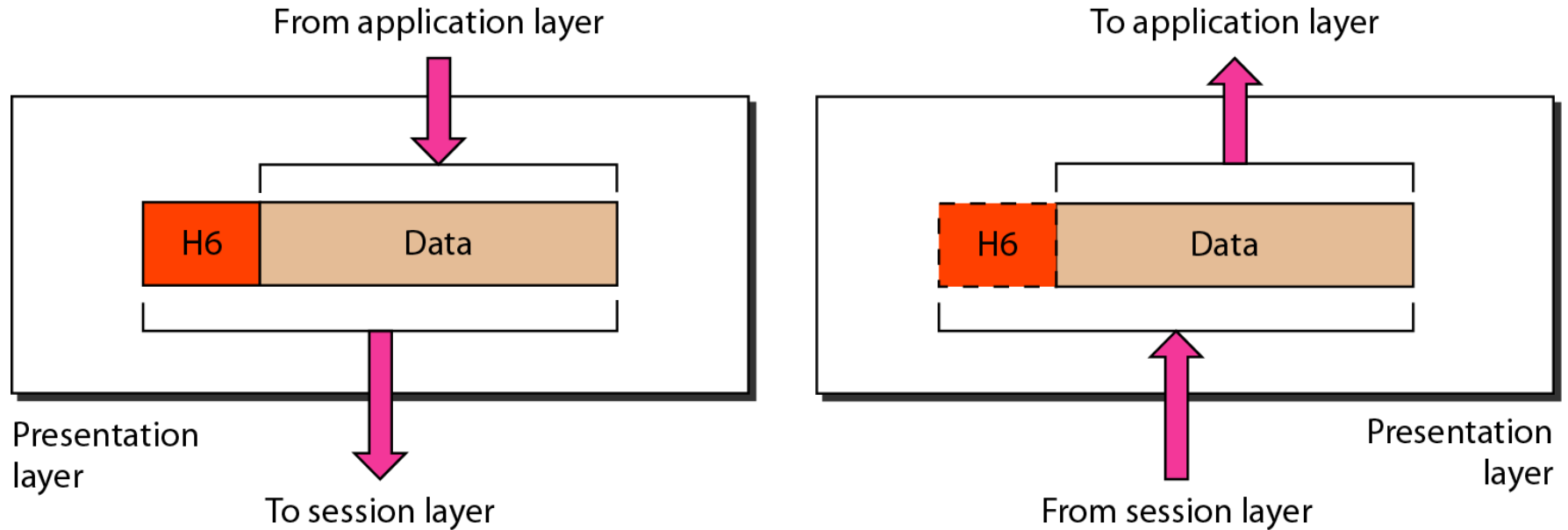


Session layer

The session layer is responsible for dialog control and synchronization.



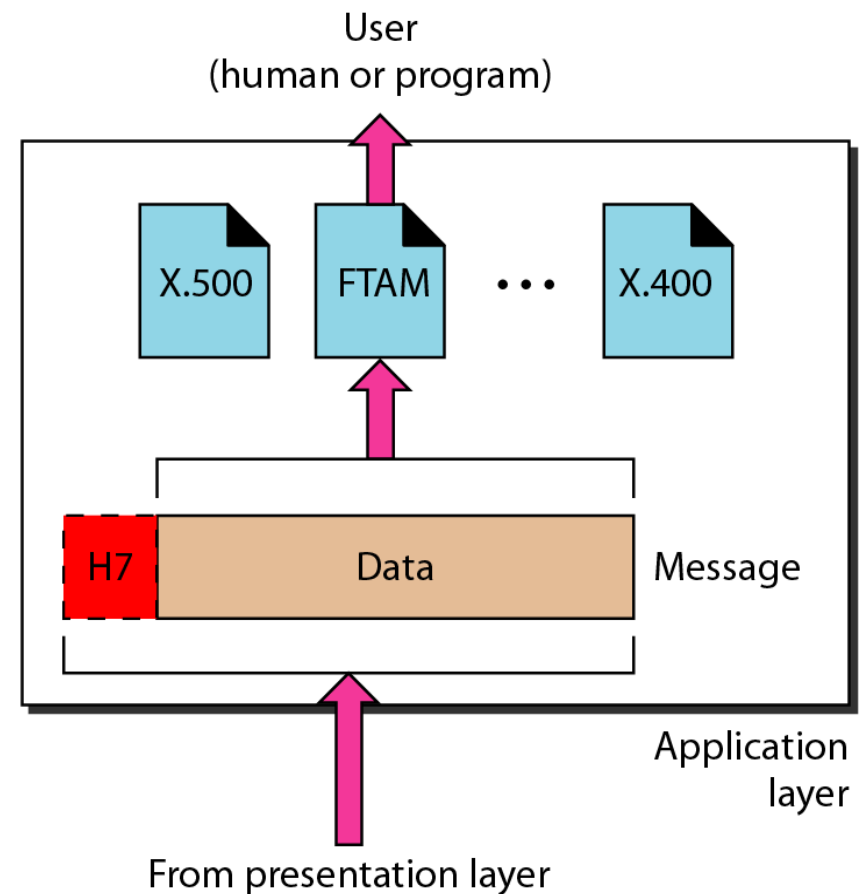
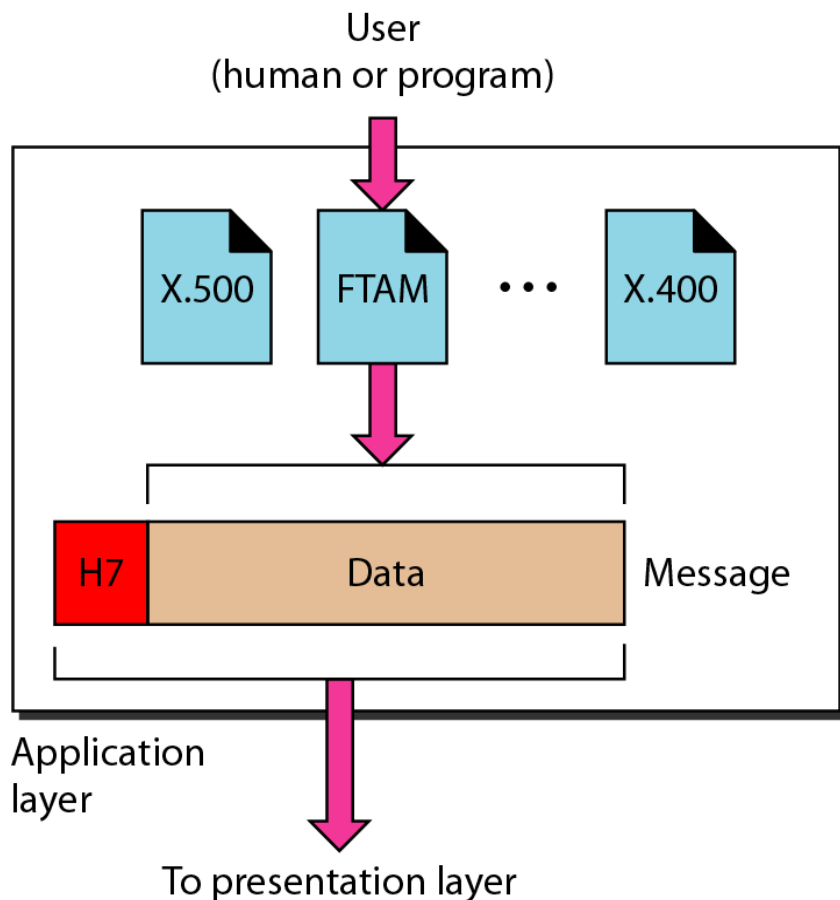
Presentation layer



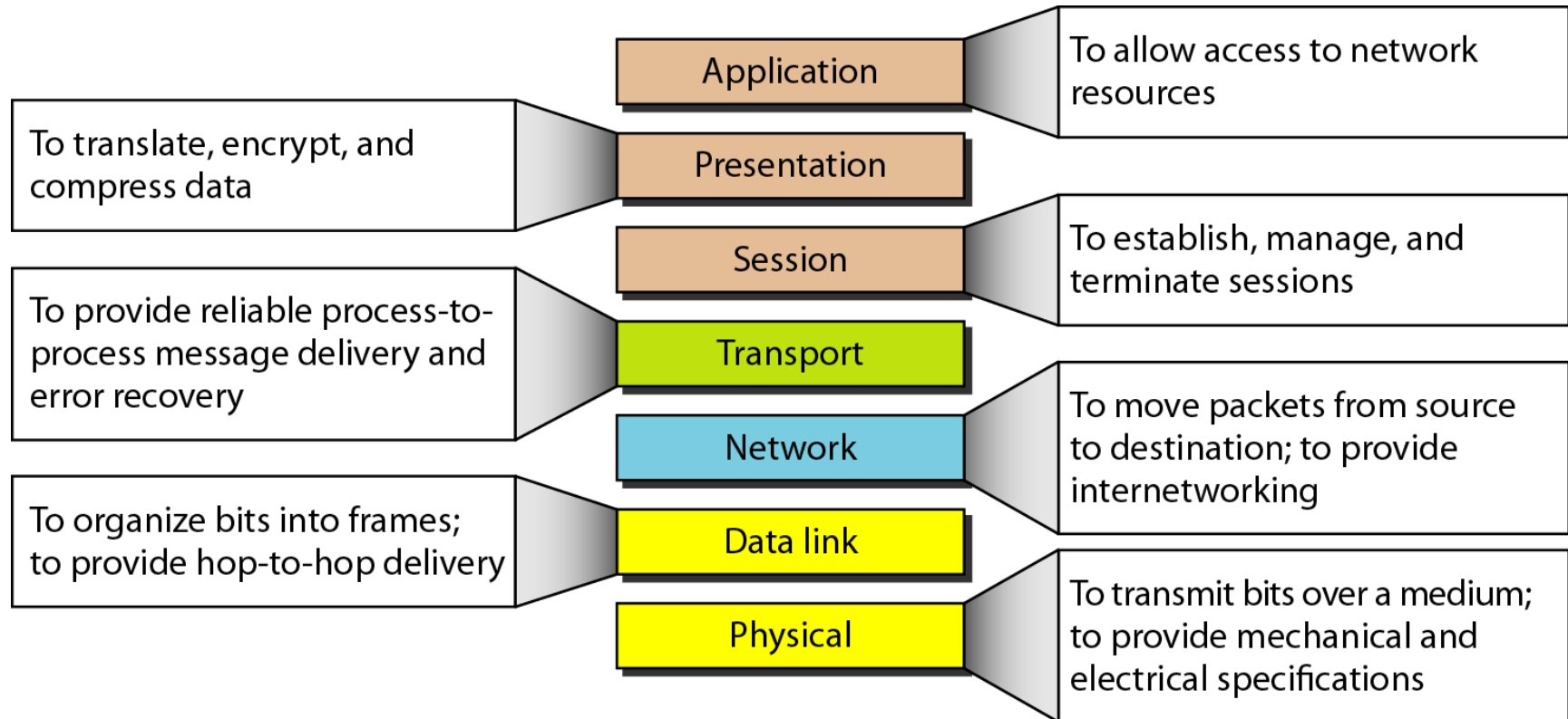
The presentation layer is responsible for translation, compression, and encryption.

Application layer

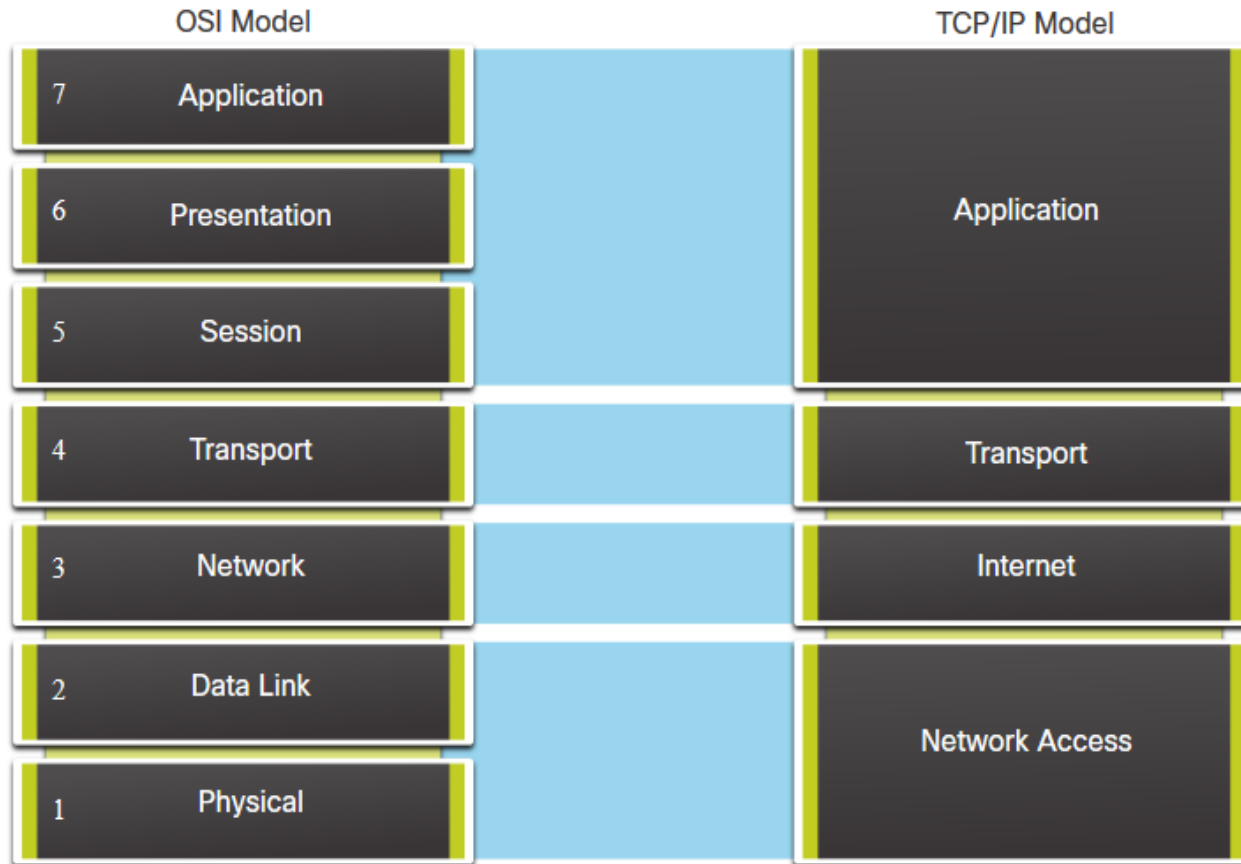
The application layer is responsible for providing services to the user.



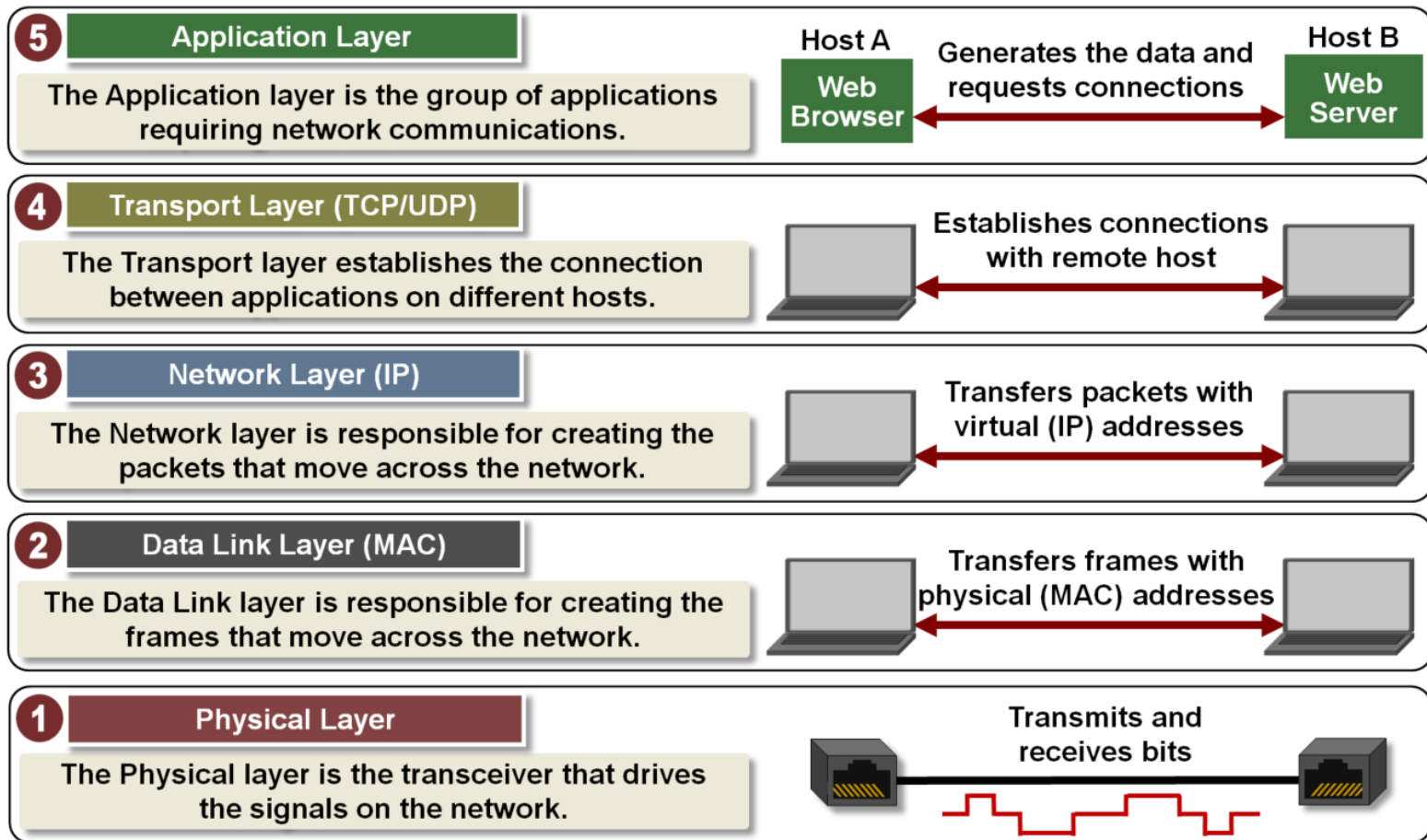
Summary of layers



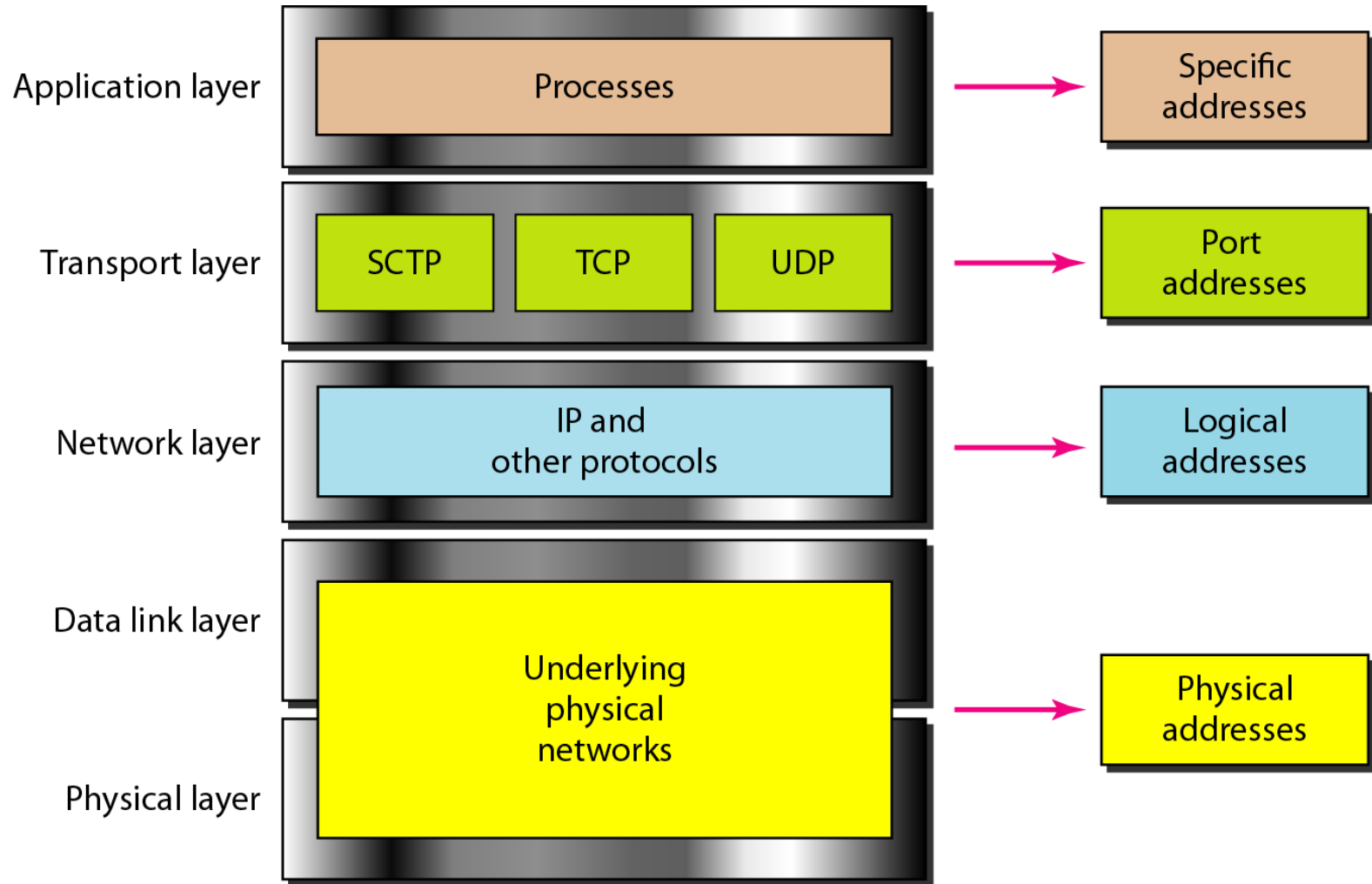
OSI and TCP/IP Model Comparison



TCP/IP Five-Layer Model



Relationship of layers and addresses in TCP/IP



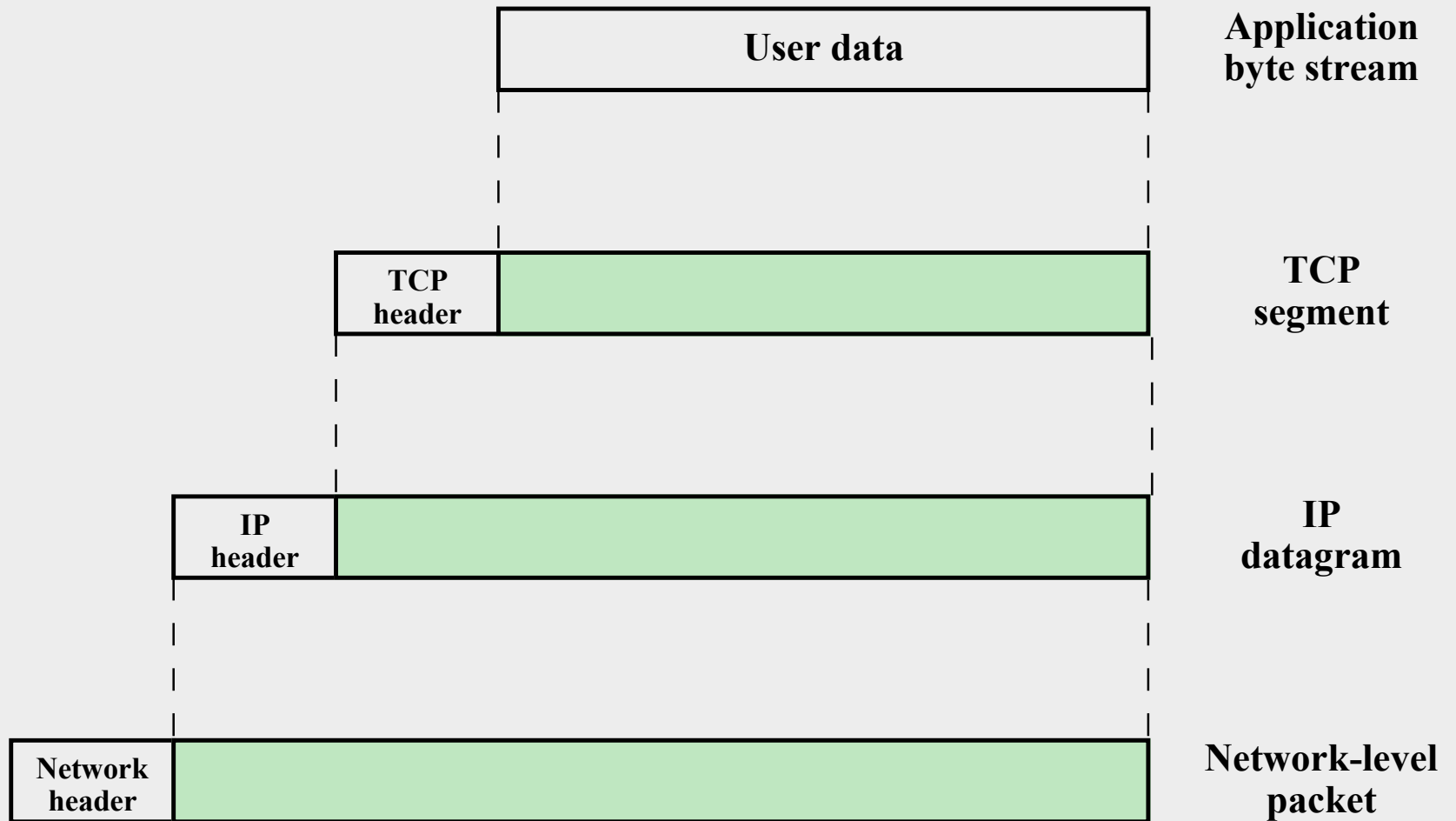


Figure 2.5 Protocol Data Units (PDUs) in the TCP/IP Architecture

Traditional Internet-Based Applications

- Three common applications that have been standardized to operate on top of TCP are:

Simple Mail Transfer Protocol (SMTP)

- Provides a mechanism for transferring messages among separate hosts

File Transfer Protocol (FTP)

- Used to send files from one system to another under user command
- Both text and binary files are accommodated

Secure Shell (SSH)

- Provides a secure remote logon capability

Table 2.2

Multimedia Terminology

Media

Refers to the form of information and includes text, still images, audio, and video.

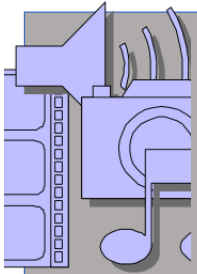
Multimedia

Human-computer interaction involving text, graphics, voice and video. Multimedia also refers to storage devices that are used to store multimedia content.

Streaming media

Refers to multimedia files, such as video clips and audio, that begin playing immediately or within seconds after it is received by a computer from the Internet or Web. Thus, the media content is consumed as it is delivered from the server rather than waiting until an entire file is downloaded.

Media Types



audio generally encompasses sounds that are produced by the human speech mechanism



image supports the communication of individual pictures, charts, or drawings



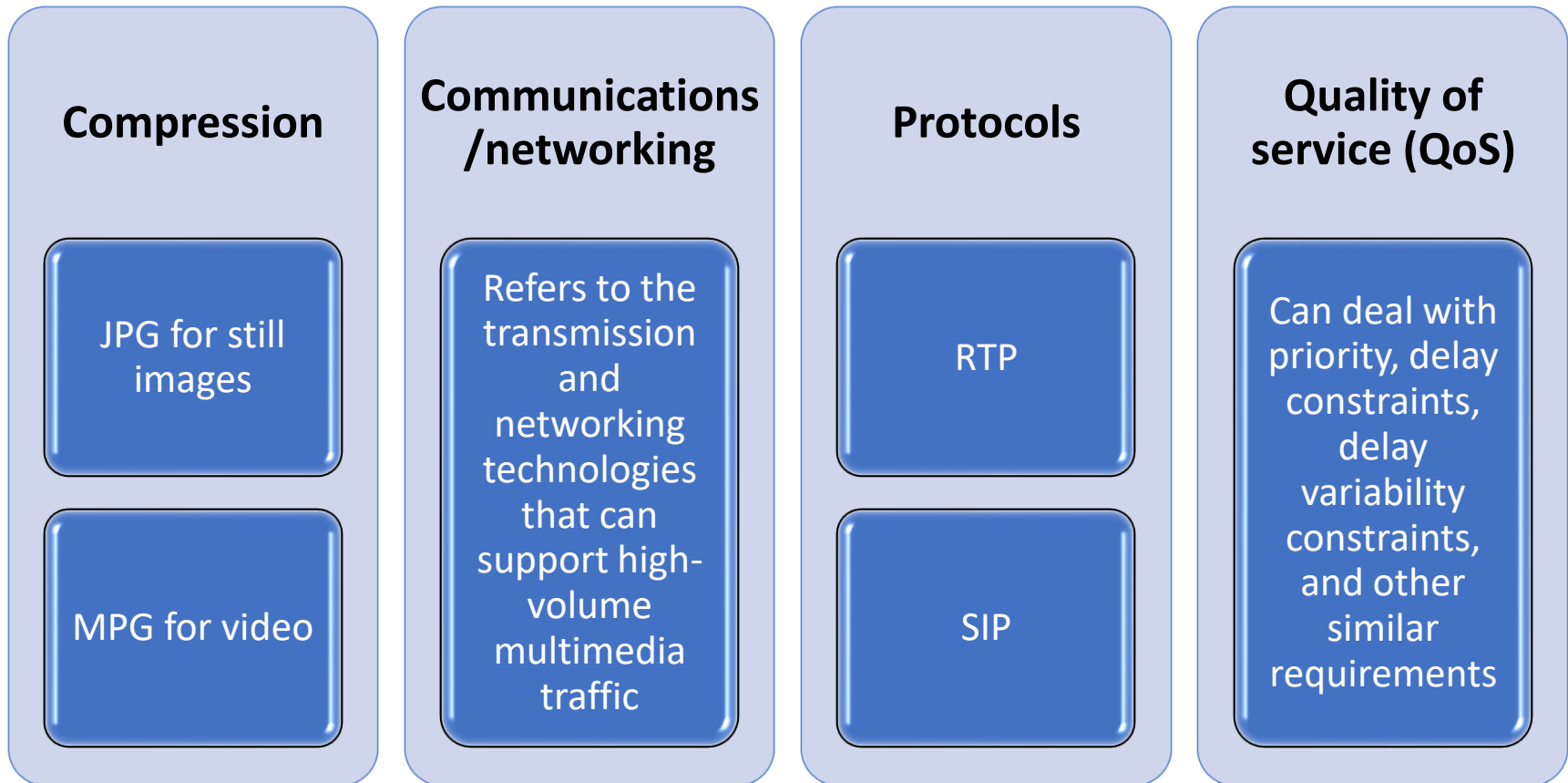
video service carries sequences of pictures in time



text is information that can be entered via a keyboard and is directly readable and printable

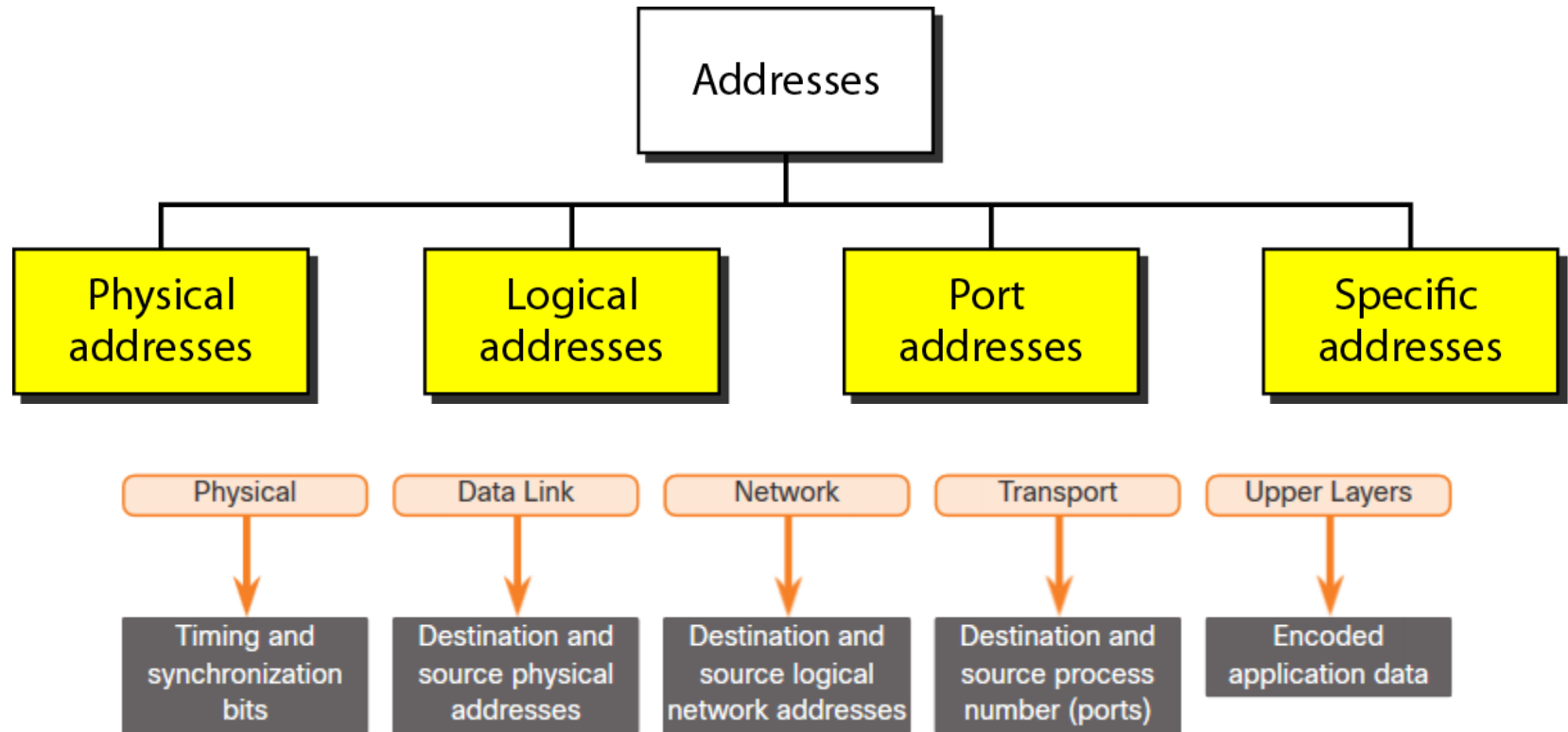
Multimedia Technologies

- Some technologies that are relevant to the support of multimedia applications are:

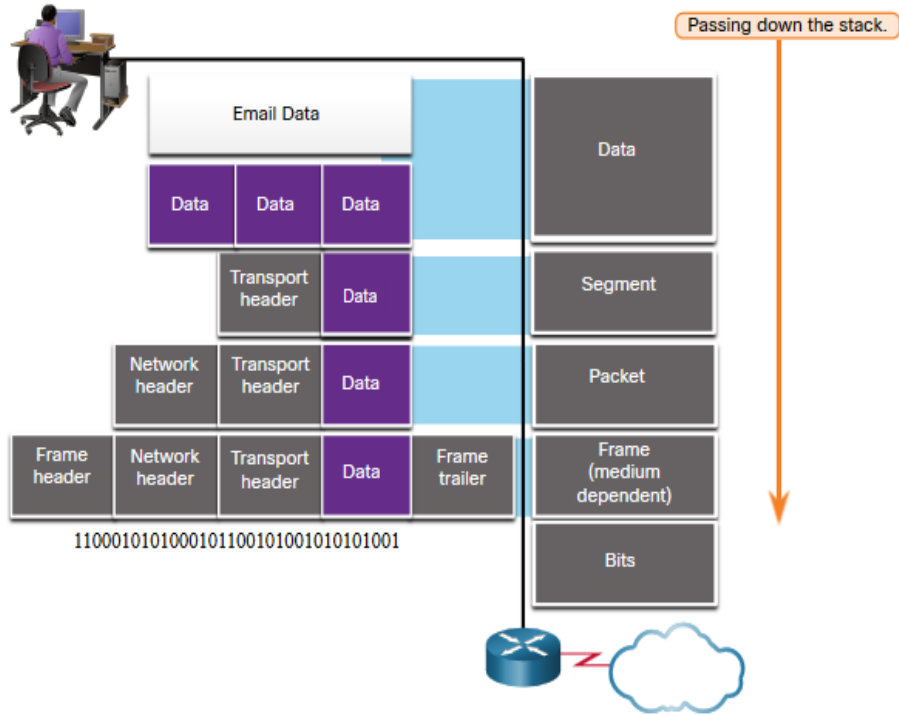


ADDRESSING

*Four levels of addresses are used in an internet employing the TCP/IP protocols: **physical**, **logical**, **port**, and **specific**.*



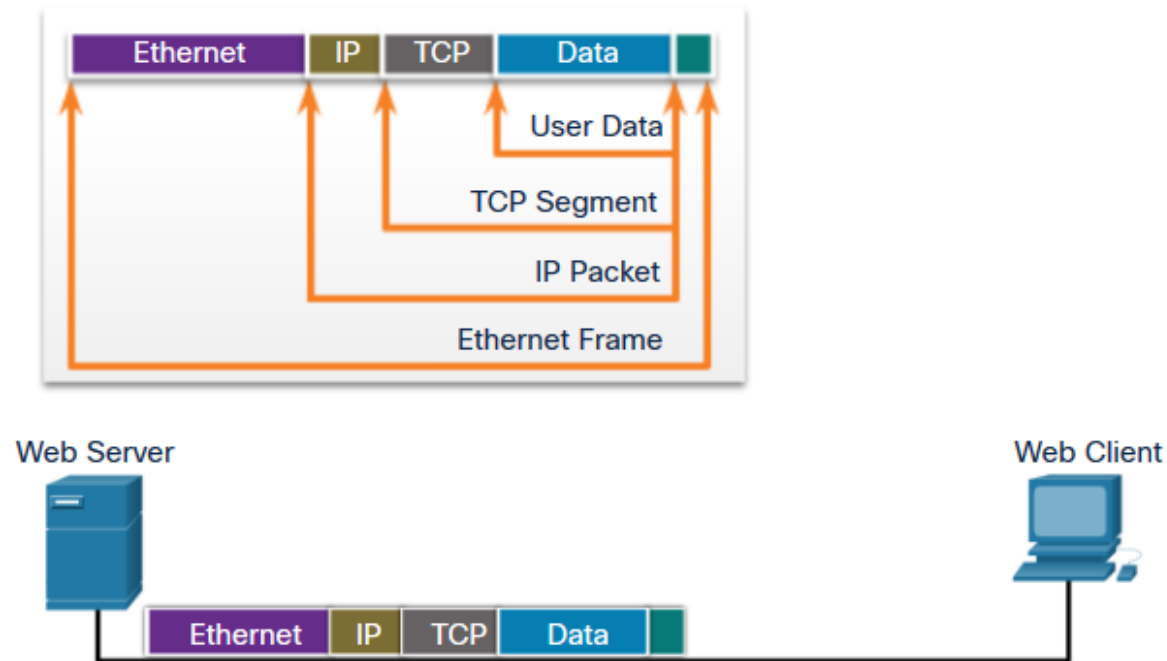
Protocol Data Units



- At each stage of the process, a PDU has a different name to reflect its new functions.
- PDUs passing down the stack are as follows:
 1. Data (Data Stream)
 2. Segment
 3. Packet
 4. Frame
 5. Bits (Bit Stream)

Encapsulation Example

- Encapsulation is a top-down process.
- The level above does its process and then passes it down to the next level of the model. This process is repeated by each layer until it is sent out as a bit stream.



De-encapsulation Example

- Data is de-encapsulated as it moves up the stack.
- When a layer completes its process, that layer strips off its header and passes it up to the next level to be processed. This is repeated at each layer until it is a data stream that the application can process.
 1. Received as Bits (Bit Stream)
 2. Frame
 3. Packet
 4. Segment
 5. Data (Data Stream)

